📘 BPSC CS Module 01: Digital Electronics & Number Systems (Desi Tricks & Sequence Notes)

> 🎯 **Target:** 15 / 15 Marks in Part-III (BPSC TRE Standard)
> **Rule:** Bina kisi faltu formula ya LaTeX code ke — Seedhi sequence, Desi tricks, aur pakke marks.

🔢 Sequence 1: Number Systems & Fast Conversion

1.1 Base aur Bit Capacity (1 Line Yaad Rakho)

- **Binary (Base 2):** Digits `0, 1` → Har digit **1 bit** leta hai.
- **Octal (Base 8):** Digits `0 se 7` → Har digit **3 bits** leta hai ($2^3 = 8$).
- **Decimal (Base 10):** Digits `0 se 9`
- **Hexadecimal (Base 16):** Digits `0 se 9` aur `A(10), B(11), C(12), D(13), E(14), F(15)` → Har digit **4 bits** leta hai ($2^4 = 16$).

1.2 Binary se Octal aur Hexadecimal Grouping Rule

```
                    [ POINT . ]
      INTEGER PART                          POINT KE BAAD
⟸⟸ (Right se Left jao)                (Left se Right jao) ⟹⟹
Point se shuru karke PEECHE chalo       Point se shuru karke AAGE chalo
(Kam pade to aage 0 lagao)              (Kam pade to peeche 0 lagao)
```

📌 **Point ke dono taraf ka simple rule:**

1. **Octal ke liye: 3-3 bits** ke jode banao.
2. **Hexadecimal ke liye: 4-4 bits** ke jode banao.
3. **Point se pehle:** Point ke theek bagal se shuru karo aur **Ulte haath (Left ←)** ki taraf chalo.
4. **Point ke baad:** Point ke theek bagal se shuru karo aur **Seedhe haath (Right →)** ki taraf chalo.

- **Example 1: $(110101.11)_2$ ko Octal me badlo**
- Point se pehle (←): `101` = **5**, `110` = **6** ⇒ Ban gaya **65**
- Point ke baad (→): `11` tha, 3 bit banane ke liye right me 0 joda → `110` = **6** ⇒ Ban gaya **.6**
- 👉 **Answer: $(65.6)_8$**
- **Example 2: $(10110111)_2$ ko Hex me badlo**
- 4-4 ka joda banao right to left:
- Pehla joda: `0111` = **7**
- Doosra joda: `1011` = 11 = **B**
- 👉 **Answer: $(B7)_{16}$**

1.3 Point wala Decimal se Binary (2 se Multiply Trick)

Agar point wala decimal number diya ho (jaise `0.625`), toh divide nahi karte — **2 se guna (multiply)** karte hain:

1. Har baar **× 2** karo.
2. Point se pehle jo number (`1` ya `0`) aaye, usko bahar nikal kar alag likh lo.
3. Bacha hua point wala hissa fir se 2 se guna karo.
4. Jab point ke baad `.00` ho jaye toh ruk jao, aur **Upar se Neeche** padh lo:

```
0.625 × 2 = 1.25  ⟶ 1 bahar nikala (bacha 0.25)
0.25  × 2 = 0.50  ⟶ 0 bahar nikala (bacha 0.50)
0.50  × 2 = 1.00  ⟶ 1 bahar nikala (bacha 0.00 → Bas yahan ruk jao!)

Upar se neeche padho: 1 → 0 → 1
👉 Answer: (0.101)₂
```

🔢 Sequence 2: 2's Complement 3-Second Inspection Trick

Pen chalane ki zaroorat nahi hai, bas aankhon se dekho:

```
          [ 1   0   1   1   0   0 ]
            |   |   |   |   └───┴── Right side ke 0s: Bilkul SAME rakho (00)
            |   |   |   └──────────── Pehla 1 jo mila: Isko bhi SAME rakho (1)
            └───┴───┴──────────────── Iske baad Left ki saari bits ko ULAT do (010)

            Result: 0 1 0 1 0 0
```

- **Step 1:** Rightmost (seedhe haath) se dekhna shuru karo.
- **Step 2:** Jab tak **pehla 1** na mile, saare 0 aur wo pehla 1 **SAME** utaar do.
- **Step 3:** Us pehle 1 ke baad baayein (left) side ki baaki saari bits ko **ULAT do (0 ko 1, 1 ko 0)**.

🔢 Sequence 3: n-Bit Range Direct Formula

Representation	Minimum se Maximum Range	Zero (0) ke roop	Total Distinct Values
Unsigned (Positive only)	**0 se +(2^n - 1)**	Sirf 1 (`00 ... 0`)	2^n
Signed Magnitude	**-(2^{n-1} - 1) se +(2^{n-1} - 1)**	2 (`+0` aur `-0`)	2^n - 1
1's Complement	**-(2^{n-1} - 1) se +(2^{n-1} - 1)**	2 (`+0` aur `-0`)	2^n - 1
2's Complement	**-2^{n-1} se +(2^{n-1} - 1)**	**Sirf 1 (`00 ... 0`)**	**2^n**

- 🎯 **8-Bit Direct Exam Numbers ($n = 8$):**
- 2's Complement range: **-128 se +127**
- Extra negative number: **-128** (`10000000`)
- MSB ka weight 2's complement me hamesha negative hota hai: **-2^{n-1}** (8-bit me `-128`).

🔢 Sequence 4: Codes (BCD, Excess-3, Gray, Hamming)

1. **BCD (8421):**

- Valid digits: **0 se 9** (`0000` se `1001`).
- Invalid states: **10 se 15** (Total 6 invalid codes hote hain).
- **Addition Rule:** Agar do BCD jodte waqt sum 9 se bada aa jaye ya carry nikle → **+6 (`0110`)** add kar do.

2. **Excess-3 (XS-3):**

- Formula: **BCD + 3 (`0011`)**.
- **Self-Complementing Code:** Excess-3 ka 1's complement nikaloge toh wo us number ka **9's complement** ban jata hai.

3. **Gray Code (Unit Distance Code):**

- Do lagatar numbers ke beech me **sirf 1 bit** ka difference hota hai.
- Shaft encoders aur error reduction me use hota hai.

4. **Hamming Code (Error Detection & Correction):**

- Error Detect karne ke liye minimum distance: **$d_{min} \geq s + 1$**
- Error Correct karne ke liye minimum distance: **$d_{min} \geq 2t + 1$**
- Parity bits (p) formula: **$2^p \geq m + p + 1$**

🔢 Sequence 5: Universal Gates Ginti Trick (1, 2, 3, 4, 5)

Dono universal gates ke liye bas ye simple sequence yaad rakho:

```
NAND Gates ki Ginti:  NOT (1) ──> AND (2) ──> OR (3) ──> XOR (4) ──> XNOR (5)
NOR Gates ki Ginti:   NOT (1) ──> OR (2)  ──> AND (3) ──> XNOR (4) ──> XOR (5)
```

Target Gate	Kitne NAND Lagenge?	Kitne NOR Lagenge?
NOT	**1**	**1**
AND	**2**	**3**
OR	**3**	**2**
NOR	**4**	**1**
NAND	**1**	**4**
XOR	**4**	**5**
XNOR	**5**	**4**

🔢 Sequence 6: Adders & Multiplexers

1. **Half Adder:**

- Sum = A ⊕ B, Carry = A · B
- Hardware: **1 XOR + 1 AND** (ya **5 NAND**)

2. **Full Adder:**

- Sum = A ⊕ B ⊕ Cin, Carry = AB + BCin + ACin
- Hardware: **2 Half Adders + 1 OR Gate** (ya **9 NAND**)

3. **Multiplexer (MUX) Formula:**

- Inputs = $2^{\text{Select Lines}}$
- **Tree MUX Formula:** 2×1 MUX ka use karke agar N×1 MUX banana ho, toh hamesha **N - 1** MUX lagenge.
- Example: 16×1 MUX banane ke liye: $16 - 1 =$ **15** MUX lagenge.

🔢 Sequence 7: Flip-Flop Table & Key Features

Flip-Flop	Inputs	Agli State (Equation)	Khaas Baat Jo BPSC Poochhta Hai
SR	S, R	Q(next) = S + R'Q	S=1, R=1 par **Invalid / Forbidden** state aati hai.
JK	J, K	Q(next) = J·Q' + K'·Q	J=1, K=1 par **Toggle** karta hai (Race Around problem).
D	D	Q(next) = D	**Data Buffer / Delay** — jo input doge wahi output aayega.
T	T	Q(next) = T ⊕ Q	T=1 par **Toggle** karta hai — **Frequency Divider (÷2)** ke roop me use hota hai.

- **Race-Around Problem:** Sirf JK Flip-Flop me aati hai jab J=1, K=1 ho.
- **Isko solve kaise karte hain? Master-Slave JK Flip-Flop** laga kar ya Edge Triggering use karke.

🎯 Top 5 BPSC Model Questions (Direct Solutions)

1. **$(110101.11)_2$ ko Octal me badlo:**

- Point ke pehle 3-3 ka group: `110 101` → 65
- Point ke baad 3 bit: `.110` → .6
- 👉 **Answer:** $(65.6)_8$

2. `10110010` **ka 2's complement kya hoga?**

- Right side se pehla 1 mila bit 2 par (`...10`).
- Baaki left side ki bits ulat do → `01001110` .
- 👉 **Answer:** `01001110`

3. **8-bit 2's complement** `11110000` **ka decimal value kya hai?**

- Pehli bit 1 hai matlab negative number hai.
- Weight: -128 + 64 + 32 + 16 = -16.
- 👉 **Answer:** `-16`

4. **XOR Gate banane ke liye kam se kam kitne NAND Gate chahiye?**

- Sequence yaad karo: NOT(1), AND(2), OR(3), XOR(4).
- 👉 **Answer:** `4`

5. **16×1 MUX banane ke liye kitne 2×1 MUX lagenge?**

- Formula: N - 1 = 16 - 1 = 15.
- 👉 **Answer:** `15`